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12 Versus 24 Hours of Postpartum Magnesium Sulfate for Seizure Prophylaxis: A Cost-Effectiveness Analysis

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Abstract

BACKGROUND: The approach to prevent eclampsia in the postpartum period utilizes magnesium sulfate but there is no evidence-based standard to guide duration.

OBJECTIVES: To assess the cost-effectiveness of an abbreviated 12-hour postpartum magnesium sulfate regimen compared with a standard 24-hour regimen.

STUDY DESIGN: A decision-analytic model was constructed to compare a 12-hour regimen with a 24-hour regimen in a theoretical cohort of 45,800 patients with preeclampsia with severe features. Probabilities, costs, and utilities were derived from the literature. Primary outcomes included incremental cost per quality-adjusted life-year (QALY), cases of eclampsia, magnesium toxicity, and maternal death. The cost-effectiveness threshold was $100,000 per QALY.

RESULTS: A 12-hour regimen in this theoretical cohort of 45,800 postpartum patients compared with a 24-hour regimen resulted in 86 more cases of eclampsia (398 vs 312) and 0.37 more deaths (10.87 vs 10.50). However, there were 656 fewer cases of magnesium toxicity (2089 vs 2745). Overall, a 12-hour regimen was a dominant strategy that resulted in decreased costs of $21.5 million and increased effectiveness of 17 QALYs.

CONCLUSION: In our study, an abbreviated duration of postpartum magnesium sulfate prophylaxis for patients with preeclampsia with severe features was a dominant strategy (lower costs, better outcomes) and cost-effective compared with the standard 24-hour regimen.

Condensation

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The authors report no conflict of interest.

A 12-hour postpartum magnesium sulfate regimen was a dominant strategy as compared to a 24-hour regimen.

Keywords

Cost-effective; postpartum; preeclampsia; eclampsia; magnesium; shorter duration

Introduction

Magnesium sulfate is used across the U.S. as the first-line agent for seizure prophylaxis for pregnant and postpartum patients with preeclampsia with severe features (1). While its use is often broadly recommended, there are potential complications including magnesium toxicity, which can lead to hypotension, respiratory distress, renal failure, or death. There are additional negative impacts that are more common, such as potential disruption in breastfeeding and newborn bonding, prolonged time to ambulation, as well as side effects of the medication such flushing, nausea, vomiting, muscle weakness, headache, dizziness, hypotension, palpitations, or tachycardia (3–13). With potential harm, choosing a regimen with the lowest effective treatment duration is key. Patient experience, safety, cost, effectiveness, and resource allocation should also be considered in all clinical decisions. Given that the risk of eclampsia is low in the postpartum period, there is a possibility that these negative impacts may outweigh the benefits of a 24-hour postpartum duration of therapy.

Given the potential negative impacts of magnesium therapy, prior studies have attempted to evaluate whether an abbreviated course of postpartum magnesium could be used (3–11). However, these studies have focused on eclampsia outcome, and given its low prevalence in this population, they have been underpowered and had little influence clinical practice (14). Instead, this study weighs the risks, costs, and health outcomes associated with duration of postpartum magnesium therapy beyond seizures. In this analysis we compared an approach of 24 versus 12 hours of magnesium therapy in postpartum patients with preeclampsia with severe features. Our hypothesis is that a 12-hour postpartum magnesium regimen is cost-effective compared to a 24-hour regimen.

Materials and Methods

A decision-analytic model was built to determine the cost-effectiveness of 12 versus 24 hours of postpartum magnesium sulfate prophylaxis strategies. Our theoretical cohort included 45,800 patients based on the annual number births in 2021 and the approximate incidence of preeclampsia with severe features in the United States (15,16). Outcomes included seizure rate, magnesium side effects, magnesium toxicity, separation from newborn, intensive care treatment, and maternal death. Model inputs were estimated from the literature (Table 1). Because we did not involve human subjects in this theoretical model, the study was deemed exempt from approval by the institutional review board. TreeAge Pro 2023 software was used for modeling.

In our model, the initial decision node stratified patients into two regimens: 12 versus 24 hours of postpartum magnesium sulfate (Figure 1). The probability of developing eclampsia was based on the Magpie trial, which observed 0.8% of patients on magnesium sulfate and 1.9% of patients not on magnesium sulfate developed eclampsia (2). The rate of seizures was based on data from studies demonstrating 70% of postpartum seizures occur between 0–12 hours postpartum while 17% occur between 12–24 hours postpartum (17). For both the 12 and 24-hour regimen, the probability of developing eclampsia was 0.55% in the first 12 hours postpartum. For those in the 12-hour regimen, the probability of having a seizure was 0.32% in the 12–24 hours while not receiving magnesium sulfate (the probability of seizure off magnesium sulfate multiplied by the seizure rate in hours 12–24) and 0.14% for the 24-hour group while continuing magnesium sulfate (the probability of seizure on magnesium sulfate multiplied by the seizure rate in hours 12–24) (2).

Health risks associated with magnesium therapy include side effects (defined as flushing, nausea, vomiting, muscle weakness, headache, dizziness, hypotension, palpitations, or tachycardia) and toxicity (defined as respiratory depression). Health risks associated with not being on magnesium included eclampsia. 24% of patients were expected to have symptoms associated with magnesium sulfate infusion (2). To prevent bias from long-term sequelae, all postpartum patients with eclampsia were assumed to make a full recovery without long-term neurologic sequalae and patients with magnesium toxicity were also assumed to recover. Given the lack of robust data, our model did not include other potential health risks associated with a longer duration of magnesium such as delayed time to ambulation and increased risks for venous thromboembolism (VTE). It was assumed that patients would be started on VTE prophylaxis protocols that likely mitigated these risks.

The probability of magnesium toxicity was derived from the MOPEP study, where intramuscular (IM) magnesium sulfate was associated with respiratory depression alone at a rate of 4.6% for a 12-hour regimen and 6.0% for a 24-hour regimen (4). The 12-hour regimen group was assumed to have a respiratory depression rate of 0% for hours 12–24. The 24-hour regimen group had a rate of 4.6% for 0–12 hours and 6% for hours 12–24. Those with loss of deep tendon reflexes (DTRs) were not included as having magnesium toxicity, as the downstream effects in our model included costs of intensive care unit (ICU) admission, which would overestimate the cost for those without respiratory depression. Thus, all patients with magnesium toxicity were assumed to need 24 hours of ICU admission for treatment of respiratory depression. We assumed that patients were unlikely to be admitted and then subsequently transferred to a lower level of care on the same day. Furthermore, some patients may need greater than 24 hours of ICU care for complications, but we preferred a conservative estimate. While eclampsia and its sequelae may require additional care, current clinical protocols from regional and national obstetric organizations do not require ICU admission as part of the standard of care (18–20).

The rate of death attributable to eclampsia was estimated from published complication rates of eclampsia/hemolysis, elevated liver enzymes, low platelet count (HELLP syndrome) from three European and one U.S. study for a combined mortality rate of 0.45% (21–24). We used an updated death rate attributable to preeclampsia of 0.02% based on 2014 U.S. data, which represents a significant reduction compared to the previous 11-year national rate of 0.09% (25).

Cost data were derived from the literature and inflated to 2023 U.S. dollars using the medical component of the Consumer Price Index. The costs of both eclampsia and magnesium sulfate treatment were taken directly from the Magpie Trial Collaborative Group (26). Specifically, we used the mean treatment cost per patient in high income countries. The main component (77%) of the treatment cost was administration of the medication, not the medication itself. Thus, the cost of administration was extrapolated from the hourly 2022 U.S. median nursing pay with the assumption that those on postpartum magnesium sulfate require 1:1 nursing (27). The need for one day of admission to the ICU was assumed for those with magnesium toxicity/respiratory depression. Adult ICU costs were derived from a large, geographically diverse, U.S. population (28). Cost of maternal death was derived from the literature (29).

We considered maternal quality-adjusted life-years (QALYs). QALYs were calculated using utilities of various health states applied to the relevant time period for that health state (life expectancies) (Table 1). Utilities are a proxy for health, with 0 representing death and 1 representing perfect health. Because there were no published utilities for several of our outcomes, we relied on similar published utilities as proxies. For example, the utility for “eclampsia” used “epilepsy with maintenance of baseline status” as a surrogate (30). Similarly, utilities for “postpartum”, “symptomatic on magnesium sulfate”, “separation from newborn”, “magnesium toxicity/respiratory depression”, and “preeclampsia with severe features” were approximated from the literature on utilities of various health care conditions (30–33). We applied utilities up to 42 hours after delivery with varied durations based on the timing of the development of magnesium toxicity and/or eclampsia. We applied utilities including eclampsia, magnesium toxicity, being on magnesium, and separation from child for 42 hours, which is beyond the duration of magnesium therapy, but not beyond the inpatient postpartum stay.

The total number of adverse clinical outcomes were compared between the 12-hour and 24-hour groups. We also calculated total costs and QALYs to determine the incremental cost-effectiveness ratio (ICER), which is the cost of the strategy per additional QALY gained. The ICER is calculated by dividing the difference in cost by the difference in QALYs between the two strategies. We set our willingness-to-pay threshold at $100,000 per QALY.

Univariate and multivariable sensitivity analyses were conducted to assess the robustness of the model. Univariate sensitivity analyses were performed based on variables identified in a Tornado diagram Monte Carlo simulation was performed using 10,000 samples to simultaneously vary model inputs. In a Monte Carlo simulation, model inputs are estimated at random within a range of prespecified distributions. In this model, probabilities and utilities (all between 0 and 1) were given beta distributions, which approximate normal distributions for values between 0 and 1. Time periods used triangular distributions and costs used gamma distributions. The result was a 95% confidence ellipse that estimated how frequently the competing strategies (12 vs 24 hours of magnesium sulfate) were optimal.

Results

In our theoretical cohort of 45,800 postpartum individuals with preeclampsia with severe features, the 12-hour and 24-hour regimens were associated with 398 and 312 cases of eclampsia, respectively (Table 2). 148 seizures occurred in hours 12–24 with the 12-hour regimen while 62 seizures occurred in hours 12–24 with the 24-hour regimen. Given the same rate of seizure in hours 0–12 (250 cases) for each group, this represents a total of 86 more instances of eclampsia with a 12-hour regimen. A 12-hour regimen also resulted in 656 fewer cases of magnesium toxicity compared with the 24-hour regimen. Importantly, there are 0.37 more annual deaths with a 12-hour regimen compared with a 24-hour regimen.

A 12-hour regimen was associated with a $21.5 million reduction in costs and 17.2 more QALYs, making it a dominant strategy (Table 2). The number needed to treat (NNT) was 531, where 531 patients need to be treated with 24 hours of magnesium sulfate (compared to 12 hours) to prevent one eclamptic seizure in this population. Approximately 7 patients out of the 531 would be expected to have magnesium toxicity.

We performed a series of univariate analyses and depicted the results using a tornado diagram, which showed that the cost of 1:1 nursing, cost of ICU care, and probability of magnesium toxicity were primary drivers of the model (Figure 2). One-way sensitivity analysis showed that the 12-hour regimen remained cost-effective when varying the cost of 1:1 nursing up to five times the baseline input. We adjusted the range in probability of magnesium toxicity down to 0.5% given that magnesium toxicity in the United States is likely much lower than what is published from countries that use IM magnesium. Thus, we varied the estimate from 0.005 to 0.07, which is lower than the high end of the range for eclampsia. In a one-way sensitivity analysis, the 12-hour magnesium strategy was dominant until a threshold probability of 0.85 for magnesium toxicity at 12 hours. Further, the 12-hour strategy remained cost-effective when varying the utility for separation from neonate up to 1, as some neonates in the cohort may remain in-room with the mother. Monte Carlo multivariable probabilistic simulation showed that the 12-hour regimen was a cost-effective strategy 89% of the time, while the 24-hour regimen was cost-effective in 11% of trials (Figure 3).

Discussion

Principal Findings: We found that the 12-hour regimen was dominant compared to the 24-hour regimen. Reducing the duration of magnesium administration resulted in both improved outcomes and decreased healthcare costs.

Results in the context of what is known: Prior studies have also shown benefits to a reduced duration of postpartum magnesium sulfate. A reduced regimen has been shown to have decreased time to ambulation (34,35), decreased time from delivery to contact with the infant (36), and lower risk of magnesium-related toxicity and other adverse events (37,38). One prior study demonstrated the cost-effectiveness of magnesium sulfate prophylaxis but failed to differentiate outcomes and costs associated with pre-delivery magnesium sulfate administration and post-delivery magnesium sulfate (26). Instead, our study investigated the efficacy of magnesium sulfate specifically during the first 24 hours postpartum, a period that has less risk of eclampsia compared to the intrapartum period.

Clinical Implications:

Current clinical practice for magnesium sulfate administration does not aim to prevent every case of eclampsia, but rather reflects a balance between the benefits of seizure prevention and the potential harms associated with treatment. In the United States, this risk threshold is implied implicitly: magnesium sulfate is routinely administered to patients with preeclampsia with severe features, while typically withheld in those without severe features (14) where the number needed to treat (NNT) to prevent one eclamptic seizure is 109 (2). In contrast, our study identified a NNT of 531 to prevent one seizure among patients with preeclampsia with severe features in the 12–24-hour postpartum window, which is even lower. In fact, we demonstrated that overall quality of life, as measured by QALYs, improved with a 12-hour regimen, despite 86 more cases of eclamptic seizures per year per 45,800 patients. Within those diagnosed with preeclampsia with severe features there is a wide range of eclampsia risk. Therefore, some patients in this category may be at greater risk of eclampsia and warrant 24 hours of postpartum magnesium, while others might not. Future studies should focus on identifying factors that predict the risk of eclampsia to support these clinical decisions.

Research Implications:

The goal of administering postpartum magnesium sulfate is to reduce seizure risk, but other potential complications should also be taken into consideration. In our study, the side effect profile and risk of magnesium toxicity were included to determine the overall impact on health outcomes and cost. We included utilities to represent the importance of patients' experience but noted a lack of targeted qualitative research in this patient population. Future research directions may include patient experience with magnesium sulfate, the influence of magnesium duration on the initiation of breastfeeding, and/or infant bonding. Better assessment of these factors will allow for more robust models and, therefore, potentially improved clinical decision-making.

Strengths and Limitations:

Our study has several strengths including the incorporation of randomized trials with detailed data regarding eclampsia risk at certain time points in the postpartum period. We included studies from high-resource clinical settings reflective of the U.S., which may derive the greatest benefit from an abbreviated postpartum magnesium course. However, when this was not available, we relied on outcomes published from low-resource settings and areas that use intramuscular magnesium sulfate instead of intravenous administration. We assumed that the toxicity of IM magnesium sulfate is similar to intravenous magnesium sulfate. The rate of magnesium toxicity may seem relatively high compared to the rate of eclampsia, but it is important to note that the risk of toxicity remains after delivery, whereas the rate of eclampsia decreases after delivery. For eclampsia at 12 hours postpartum, we evaluated the robustness of our results across a range of prevalence from 0.003 to 0.008. For magnesium toxicity we varied the estimate from 0.005 to 0.07, which is lower than the high end of the range for eclampsia. There are also several methodological limitations inherent in cost-effectiveness methodology, including a certain degree of uncertainty in the inputs for the selected probabilities, costs, and utilities. The incidence of magnesium toxicity was limited as many studies combine both respiratory depression and loss of DTR, which are associated with different clinical consequences. These outcomes were isolated when supplementary data were available. However, we were reassured that even large variations in the inputs in our sensitivity analysis did not affect the study's findings. We could not incorporate several potentially important clinical scenarios where there could be risks and costs of magnesium that are institution-specific. For example, the cost associated with nursing protocols, the potential for harm from medication errors, policies where neonates are not allowed to room in with patients on magnesium and the subsequent effects of those policies on bonding, breastfeeding initiation and success, as well as patient satisfaction. However, regarding nurse-to-patient ratios for patients on magnesium, we feel that our sensitivity analysis range for nursing costs covers possible variations in care, such as a 1:2 ratio for example. Finally, we did not evaluate the cost effectiveness of magnesium in the setting of preeclampsia without severe features.

Conclusions:

While the 24-hour regimen has been a longstanding clinical recommendation, our data suggest benefit to a reduced duration. This highlights an opportunity for patient-centered approaches to determine timing of magnesium cessation. This may be particularly important in cases where newborn visitation on another unit is limited by magnesium protocols. Ideally this would be accompanied by an individualized eclampsia prediction tool accounting for factors beyond postpartum timing to include blood pressure status, neurological symptoms, lab abnormalities, and diuresis (14,39). In the meantime, we believe these findings can be used to allow for individualization of policies for duration of postpartum magnesium prophylaxis.

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AJOG at a Glance

- A. The aim of the study was to assess the cost-effectiveness of a 12-hour postpartum magnesium sulfate regimen compared to a 24-hour regimen.
- B. The key finding is that a 12-hour regimen was a dominant strategy that resulted in $21.5 million savings with an increased effectiveness of 17 QALYs.
- C. It has been proposed that a reduced duration of postpartum magnesium sulfate prophylaxis is safe. This is the first study to determine the cost-effectiveness of a reduced treatment duration.

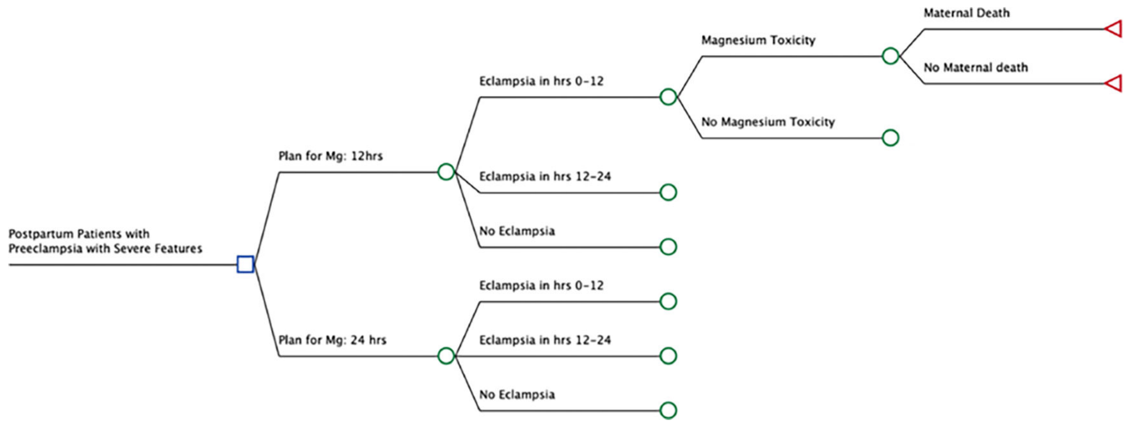

Fig. 1.

Schematic tree created using TreeAge software. Mg, magnesium.

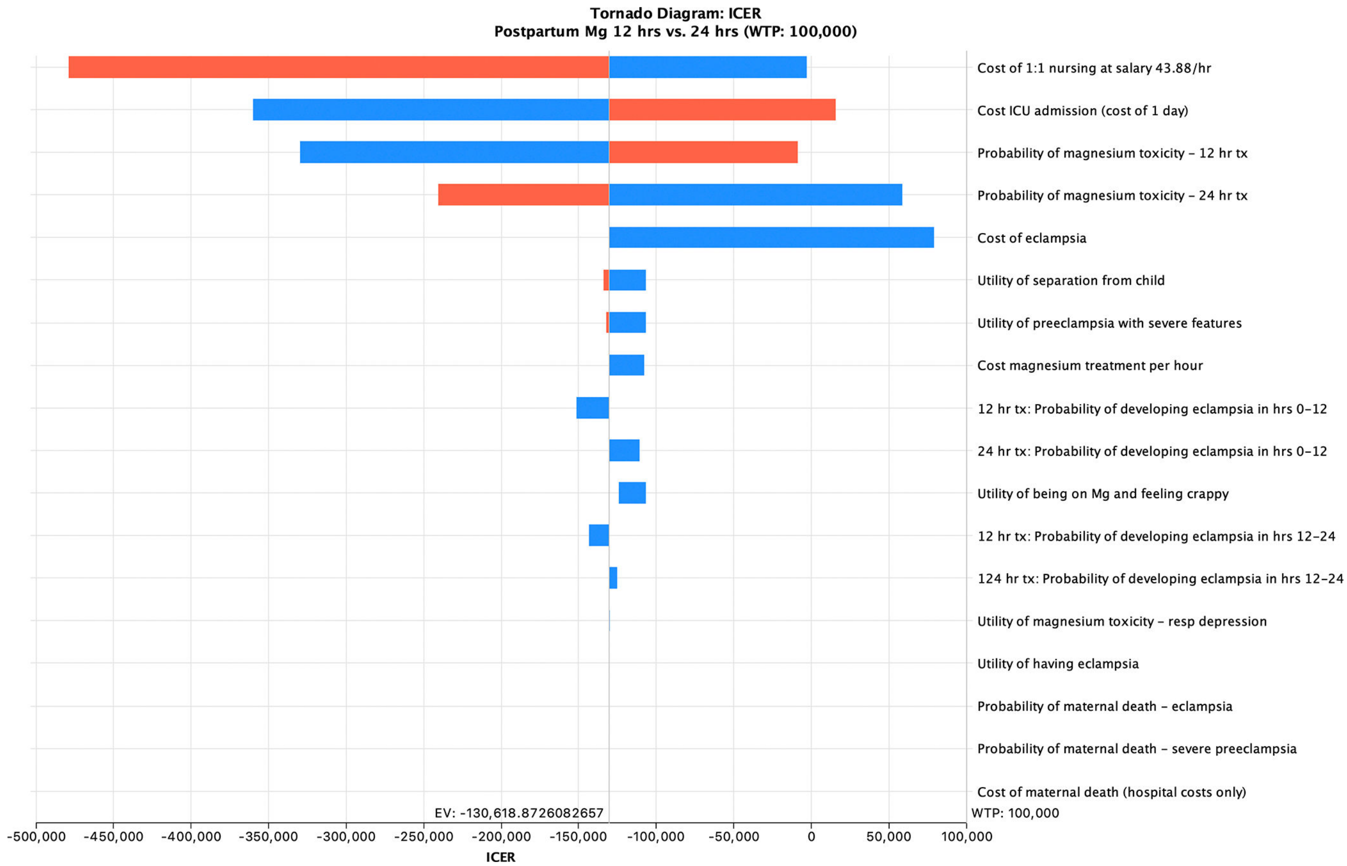

Fig. 2.

Tornado diagram created using TreeAge software. Incremental cost-effectiveness ratio to a 12-hour regimen vs 24-hour regimen (willingness to pay [WTP]: 100,000). ICU, intensive care unit; EV, equal value.

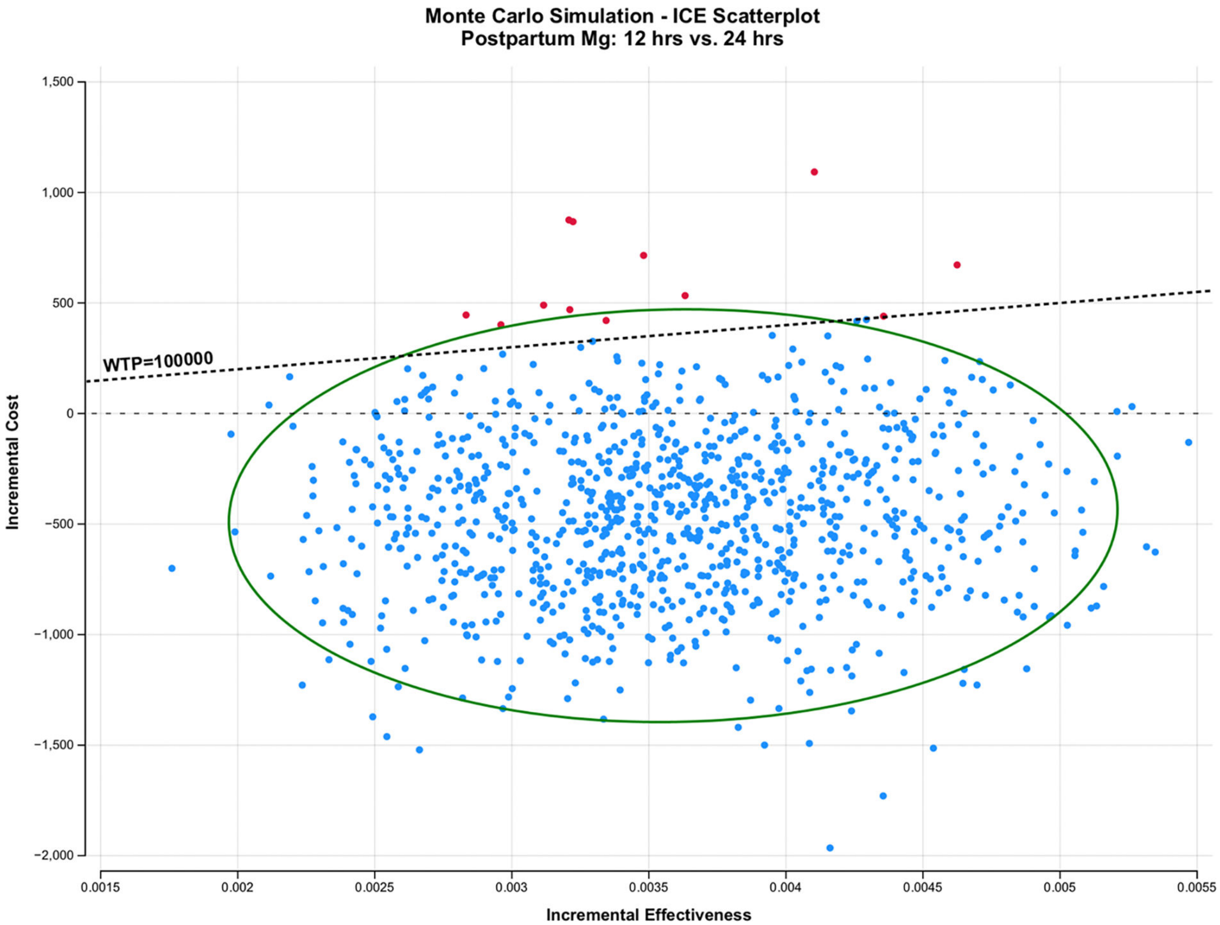

Fig. 3.

Monte Carlo simulation created using TreeAge software. ICE, incremental cost-effectiveness; WTP, willingness to pay.

Table 1.

Probabilities, Utilities, and Notable Costs Used in the Model

Variable	Value	Range in Sensitivity Analysis	References
Probabilities			
Probability of developing eclampsia on Mg	0.008	0.004 – 0.112	2
Probability of developing eclampsia placebo	0.019	0.013 – 0.025	2
Probability of maternal death - eclampsia	0.004	0.002 – 0.01	18–21
Probability of maternal death - severe preeclampsia	0.0002	0.0001 – 0.00026	22
Probability of magnesium toxicity - 12	0.046	0.005 – 0.07	4
Probability of magnesium toxicity - 24	0.06	0.005 – 0.09	4
Postpartum seizure: yes, hours 0–12	0.69	0.5 – 0.9	17
Postpartum seizure: yes, hours 12–24	0.17	0.02 – 0.3	17
12: Probability of developing eclampsia 0–12	0.0054	0.003 – 0.008	
12: Probability of developing eclampsia 12–24	0.0032	0.002 – 0.005	
24: Probability of developing eclampsia 0–12	0.0054	0.003 – 0.008	
24: Probability of developing eclampsia 12–24	0.0014	0.0003 – 0.002	
Costs			
Cost magnesium treatment 12	$86.41	20 – 150	23
Cost magnesium treatment 24	$172.82	40 – 300	23
Cost magnesium treatment per hour	$7.14	2 – 13	23
Cost of ICU admission	$18,304.40	4,300 – 32,300	25
Cost of 1:1 nursing at salary	$43.88/hr	10 – 80	24
Cost of eclampsia	$13,200.37	3,100 – 23,300	23
Maternal death	$9,232.62	2,200 – 16,300	26
Utilities			
Utility of having eclampsia	0.66	0.62 – 0.7	28
Utility of magnesium toxicity	0.7	0.66 – 0.74	28, 29
Utility of preeclampsia with severe features	0.83	0.56 – 1.0	30
Utility of being on Mg	0.842	0.81 – 0.88	28
Utility of separation from child	0.87	0.85 – 0.89	27
Utility of being postpartum	0.99	0.98 – 0.99	27

Table 2.

Outcomes associated with 12 versus 24-hour regimen with a theoretical cohort of 45,800

Variable	12 hours	24 hours	Difference
Maternal death	10.87	10.50	0.37
Eclampsia	398	312	86
Eclampsia 0–12 hours	250	250	0
Eclampsia 12–24 hours	148	62	86
Magnesium toxicity	2089	2745	−656
Cost (USD)	$69,415,745	$90,980,517	−$21,564,772
Effectiveness (QALYs)	255	238	17